

Eren Badur

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EDUCATION

Sabancı University, Istanbul/Turkey

Fall 2021 – Spring 2025

- Computer Science & Engineering BSc
- Faculty of Engineering and Natural Sciences
- Dean's List, High Honor - Spring 2025
- Dean's List, Honor - Fall 2024
- Graduated in June 2025

Robert College, Istanbul/Turkey

Fall 2016 – Spring 2021

- Graduated in June 2021

WORK EXPERIENCE

Riskoptima Wealth Tech, Financial Engineer, Istanbul, Turkey

July 2025 – January 2026

- Developed 2 and 4-legged short iron condors for 0-DTE and 1-DTE options volatility trading strategies with Python.
- Discretized the Heston model using the Euler-Maruyama method and Bergomi Rough Volatility method.
- Optimized the spread between the long and short put legs A and B with nested-Monte Carlo simulations.

QNB, Machine Learning Intern, Istanbul, Turkey

July 2024 – September 2024

- Worked with imbalanced datasets on customer payment type, credit card retention, and instant loan application.
- Implemented logistic regression, decision tree, random forest, XGBoost, LightGBM, and ensemble models, performed data cleaning, outlier detection, and feature engineering. Used ensemble models, binning, oversampling, and undersampling techniques to handle imbalanced data.
- Created successful models that obtained high ROC AUC and F1 scores.

PROJECTS

BTCUSDT Technical Analysis Trading Bot in C++

August 2025 – Present

- Developed a rule-based trading bot as a systemctl service that opens a long or short position for BTCUSDT in Binance Futures, using technical indicators such as RSI, ATR, EMA, MACD, VPOC, and the Ichimochi cloud in an Ubuntu cloud server.
- Trained a Hidden Markov Model with Viterbi Approximation for market regime detection. Trained an XGBoost model in C++ to rule out positions when probability is lower than the meta threshold.
- Obtained a profit of 20% and above in USD for multiple combinations in 1-year backtest between 1 January 2025 and 1 January 2026.

ENS492, Graduation Project, Müdek Document Tracking System

Fall 2024 & Spring 2025

- Developed a web-based system document management platform for MÜDEK and YÖK compliance, enabling professors and assistants to securely upload necessary documents at the end of each term.
- Deployed to the university server, scheduled for institutional use starting next semester.
- Implemented the backend in PHP, frontend with HTML, Tailwind CSS, and used MariaDB for database management.
- Integrated Sabancı CAS Login for secure, university-authenticated access and privacy control.

CS445, Natural Language Processing, Stance Detection on Political Tweets**Fall 2024**

- Used a dataset from an older study consisting of real tweets about Biden and Trump before the 2020 US elections.
- Surpassed the original study in terms of model performance. Used BERT, RoBERTa, PoliBERTweet transformers; executed text cleaning, feature extraction, feature engineering. Used log-odds ratio with Dirichlet priors to identify significant stance-related tokens for each candidate.

CS308, Software Engineering, E-Commerce Platform**Fall 2024**

- Developed an e-commerce site with NodeJS and ReactJS. MongoDB was used as database.
- Implemented security features such as password hashing and authorized navigation.

CS412, Machine Learning, Influencer Category and Popularity Detection**Fall 2024**

- Developed a multi-class classification model to categorize Instagram influencers and a regression model to predict content popularity.
- Used TF-IDF and BERT for feature extraction. LightGBM emerged as the best-performing model across rounds, enhanced with hyperparameter tuning and PCA for efficiency.

CS408, Computer Networks, E-Commerce Platform Spring 2023

- Developed a rock, paper, and scissors server-client game that allows up to 10 people to play at the same time using C# and .net framework.

CS310, Mobile Application Development, Rate My Professors Sabancı**Fall 2023**

- Designed a mobile application called “Rate My Professors Sabancı” that allows users to express their opinions and give recommendations about courses and professors at Sabancı University.
- Used Java, Spring Tool Suite, Android Studio, MongoDB, and Docker.

SKILLS

Languages:

- English (Advanced), German (Intermediate), Turkish (Native)

Computer Science:

- Programming Languages: C++ (Fluent), Python (Fluent), SQL (Fluent), Bash (Intermediate), C# (Intermediate), Java (Intermediate), JavaScript (Intermediate), TypeScript (Intermediate), PHP (Basic), R (Basic)
- Experienced in full-stack development, supervised learning, unsupervised learning, and reinforcement learning
- Experience with PyTorch, Hugging Face transformers, neural networks, Keras, and TensorFlow
- Experience with Docker, MongoDB, MySQL, MariaDB, GitHub in Scrum projects

ACHIEVEMENTS

- First place in the National Exam (TEOG) in 2016.
- Second place in Europe in Mathematics Without Borders Tournament in 2015.
- Leading player in SEV Schools Chess team between the years 2010 and 2014.
- First place in Üsküdar Regional Middle Schools Chess Tournament in 2010.
- Second place in Turkey in Abalone tournament in 2013.
- Coordinator at IEEE Sabancı Club for five months, the school’s most prestigious engineering student club.